Dr. Arachchige Jayaruwani Fernando

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Professional Summary

An accomplished and passionate Senior Lecturer in Agricultural Engineering with over 13 years of experience driving excellence in teaching, research, and academic leadership. Expertise spans Agricultural Machinery, Mechanization, and Postharvest Technology, underpinned by a strong technical foundation in engineering design, HVAC systems, and sensor-based controls.

- Research & Innovation: Doctoral research focused on optimizing heat pump drying systems, leading to practical agricultural innovations. Co-inventor of a patented farm machine currently recommended for national recognition.
- Awards & Recognition: Recipient of nearly 20 research excellence awards, including the Outstanding Doctoral Research Award – Gold Medal for the Board of Study in Agricultural Engineering, Postgraduate Institute of Agriculture, University of Peradeniya and Vice Chancellor's Award for the Most Outstanding Young Researcher, Rajarata University of Sri Lanka. Internationally honored through induction into Gamma Sigma Delta, Alpha Epsilon, and Epsilon Pi Tau honor societies for academic and professional excellence, and accredited by the Staff and Educational Development Association (SEDA), UK, for outstanding teaching and professional development.
- Teaching & Mentorship: Proven success in curriculum development and in mentoring undergraduate and postgraduate students, effectively integrating cutting-edge research into pedagogy to foster experiential learning.
- Academic Contribution: Strong record of high-impact publications and international collaborations, consistently bridging theory and practice to deliver tangible agricultural solutions.

Areas of Expertise

Technical & Research Expertise:
Process Optimization in Food & Agricultural Engineering — Farm Machinery & Mechanization — Drying and Dehydration Technologies — HVAC Systems for Controlled Environment Agriculture — LED Applications for Postharvest Storage — Lean Manufacturing — Data Analysis & Experimental Design — Scientific Writing & Peer-Reviewed Publications

Academic & Professional Competencies:
Curriculum Development & Pedagogical Innovation — Research Supervision & Mentoring — Academic Quality Assurance & Curriculum Review — Peer Review for International Journals — Classroom Management & Lesson Planning — Learning Management Systems — International Collaboration & Affairs — Academic Event Organization & Outreach

Educational Qualifications

2022 **Doctor of Philosophy (PhD)** in Agricultural and Biosystems Engineering, University of Peradeniya, Sri Lanka
Thesis title: Optimization of heat pump drying by real-time control of air flow through cascade evaporators and parallel-flow condensers.

2017 **Master of Science (MS)** in Agricultural and Biosystems Engineering, Iowa State University, USA
Major: Industrial and Agricultural Technology

2015 **Master of Philosophy (MPhil)** in Agricultural and Biosystems Engineering, University of Peradeniya, Sri Lanka
Thesis title: Drying and roasting characteristics of chili (Capsicum annuum L.) exposed to different wavelengths of far-infrared radiation.

2011 **Bachelor of Science (BSc) in Agriculture**, Rajarata University of Sri Lanka
Major: Agricultural Engineering – Second Class Upper Division.
Thesis title: Design and development of a two-wheel tractor-driven coconut fertilizer applicator.

2010 **Diploma in English**, Rajarata University of Sri Lanka – One-year program

Career Experience

Senior Lecturer Sri Lanka
Rajarata University of Sri Lanka 2017 upto now
Undergraduate courses: Design Philosophy, Food Process Engineering, Engineering Physics, Innovation & Product Development

Lecturer Sri Lanka
Rajarata University of Sri Lanka 2012–2017
Courses: Food Process Engineering, Food Packaging, Basic Engineering Physics, Design Philosophy, Energy Production and Management, Electronics in Agriculture, Mathematics.

Teaching Assistant (Demonstrator) Sri Lanka
Rajarata University of Sri Lanka (RUSL) 2011–2012
Responsibilities: Assessing course assignments and conducting practical sessions.

Additional Experience

Visiting Lecturer 2025
PGIA, University of Peradeniya
Conduct 4 hour lecture on 'Drying' for M.Sc. in Agricultural & Biosystems Engineering students.

Master of Agriculture, RUSL Sri Lanka
Postgraduate course: Advanced Machinery Systems Engineering – Engineering Design. 2020–2022

Visiting Lecturer Sri Lanka
Aquinas College of Higher Studies, Colombo 2020–2021
Courses: Farm Power Mechanization and Irrigation & Water Resource Engineering.

Comprehensive Panel Member Sri Lanka
Board of Study in Agriculture, RUSL 2022–2024

Graduate Research Assistant United States
Department of Agricultural & Biosystems Engineering 2015–2017
Iowa State University, Ames, Iowa
Research on Food Security and Lean Management.

Technical Proficiency

Python Programming, LaTeX, AutoCAD, Arduino Programming, Microsoft Office

Professional Online Profiles

Google Scholar – Jayaruwani Fernando

ORCID – 0000-0003-2292-547X

LinkedIn – Jayaruwani Fernando

Scopus – 57224782498

Web of Science ResearcherID – AAT-4261-2021

Awards & Honors

National and International Awards

- **Merit Award, Presidential Awards for Inventions (2021/2022).** 2025 (*to be awarded*)
- **Best Scientific Paper Award.** 2025
  Special Session on Next-Generation Technologies for Sustainable, Resilient, and Smart Food Systems, The Kandy Conference (ICSBE & Next-Gen Innovation).
- **Outstanding Doctoral Research Award – Gold Medal.** 2025
  Board of Study in Agricultural Engineering, Postgraduate Institute of Agriculture, University of Peradeniya.
- **Grain Elevator and Processing Society (GEAPS) Iowa Chapter Scholarship.** 2017
  Awarded to support education and research in grain handling, processing, and storage.

Academic Honors and Distinctions

- **Elected to Gamma Sigma Delta – The Honor Society of Agriculture.** 2017
  Recognized for outstanding scholarship, leadership, and service in agriculture.
- **Elected to Alpha Epsilon – The Honor Society of Agricultural and Biological Systems Engineering.** 2017
  Recognized for academic excellence and professional contributions to the field.
- **Elected to Epsilon Pi Tau – The International Honor Society for Professions in Technology.** 2017
  Recognized for outstanding scholarship, leadership, and service in technology-related professions.

Institutional Awards for Research Excellence

Rajarata University of Sri Lanka (RUSL)

- **Vice Chancellor's Award – Researcher with the Second-Highest *h*-Index, Faculty of Agriculture.** 2025
  Based on citations accumulated during 2022–2024.
- **Vice Chancellor's Award – Researcher with the Third-Highest Number of Citations, Faculty of Agriculture.** 2025
  Based on citation counts for 2022–2024.
- **Vice Chancellor's Award – Researcher with the Highest Number of Citations, Faculty of Agriculture.** 2024
  Secured first place based on citations accumulated during 2021–2023.
- **Vice Chancellor's Award – Researcher with the Highest *h*-Index, Faculty of Agriculture.** 2024
  Secured first place based on *h*-index calculated over 2021–2023.
- **Vice Chancellor's Award – Researcher with the Highest Number of Indexed Journal Articles, Faculty of Agriculture.** 2023
  Secured first place for Web of Science–indexed journal publications during 2020–2022.
- **Vice Chancellor's Award – Most Outstanding Young Researcher, Faculty of Agriculture.** 2022
  Awarded for exceptional academic performance, including at least three original research publications in indexed journals within the preceding three years.
- **Vice Chancellor's Award – Researcher with the Highest Number of Indexed Journal Articles, Faculty of Agriculture.** 2022
  Secured first place for Web of Science–indexed publications during 2019–2021.
- **Excellence Award – Top Ten Researchers with the Highest Number of Indexed Journal Articles, University-wide.** 2022
  Ranked fifth among all RUSL faculty members based on Web of Science–indexed publications during 2020–2022.

- **Certificate and Award of Excellence – First Place in Research, Scholarly, and Creative Work, Faculty of Agriculture.** 2021
- **Top Twenty Researcher Award.** 2021
  Awarded to the top 20 faculty members with the highest research publication output in 2020.
- **Vice Chancellor's Award – Annual Publication Recognition.** 2024, 2023, 2022, 2021
  In recognition of publishing peer-reviewed research articles in indexed journals during 2023, 2022, 2021, and 2020, respectively.

Teaching and Presentation Awards

- **Teacher Accreditation, Staff and Educational Development Association (SEDA), United Kingdom.** 2014
  Accredited for excellence in teaching, professional development, and commitment to high-quality higher education.
- **Most Outstanding Presenter – Agricultural Engineering (Gold Medal and Certificate).** 2013
  Third Annual Research Symposium, RUSL.
- **Most Outstanding Presenter – Agricultural Engineering.** 2011
  Undergraduate Research Symposium, RUSL.

Professional Memberships

- American Society of Agricultural and Biological Engineers
- National Science Foundation, Sri Lanka
- Organization for Women in Science for the Developing World

Training and Certifications

Pedagogical and Academic Development

- Certificate in Teaching in Higher Education (CTHE), University of Colombo, Sri Lanka (2014).
- Four-day intensive training programs on Outcome-Based Education and Learner-Centered Teaching
  University of Peradeniya, Sri Lanka (2021); University of Colombo, Sri Lanka (2021); Wayamba University of Sri Lanka (2020)
- Three-day training for Career Guidance Unit Staff, organized by the Standing Committee on Career Guidance, University Grants Commission, in collaboration with HETC (2014)

Programming and Data Analytics

- Python Programming Certificate, University of California San Diego, USA
  Introduction to Programming (2020); Python Programming Fundamentals (2022); Intermediate Python (2024); Data Analytics Using Python (2025)

Technical and Industry Training

- Two-day training program on *Introduction to Agricultural Machines to Minimize Farm Cost*, National Engineering Research and Development Centre (NERDC), Sri Lanka, October 10–11, 2025
- Applications of Machine Learning Techniques in Agricultural Engineering 5th CIGR International Conference on Integrating Agriculture and Society through Engineering (2021)
- Grain Elevator and Processing Society, USA (2017)
  Grain Quality Management; Aeration Systems Design and Fan Operation; Grain Drying
- Sri Lanka–West German Farm Mechanization Project, Farm Mechanization Training Center, Anuradhapura (2009–2010)
  Operation and Maintenance of Farm Machinery; Operation and Maintenance of Two-Wheel and Four-Wheel Tractors; Crop Cultivation under Micro-Irrigation

Research Training

- Research trainee, National Engineering Research and Development Centre (six-month undergraduate research project), 2011

Research Supervision

Postgraduate: Serving as a member of the Supervisory Committee for Ms. C. A. Kumarasiriwardhana (PGIA-18060), M.Sc. (Coursework & Research) in Agricultural and Biosystems Engineering, Postgraduate Institute of Agriculture, University of Peradeniya. The thesis is titled Machine Learning-Based Predictive Modeling of Drying Kinetics and Quality Attributes of Turmeric (*Curcuma longa* L.) under Hot-Air and Microwave Drying.

Undergraduate: Supervised and co-supervised 23 six-month undergraduate research projects from 2013–2025 in agricultural mechanization, postharvest process engineering, and IoT-enabled control systems.

- Innovative infrared paddy dryer prototype: Design, development, and performance assessment – 2025
- Real-time IoT-based performance monitoring and control system for infrared paddy drying – 2025
- Development of an Artificial Intelligence-based automated shrimp size estimation model using image processing for Pacific White Shrimp (*Litopenaeus vannamei*) Culture Systems in Sri Lanka – 2025

- Effect of different wavelengths of light-emitting diodes on harvested water spinach (*Ipomoea aquatica* Forsk.) – 2024
- Drying characteristics and quality of paddy exposed to infrared radiation – 2024
- Performance evaluation of an industrial heat pump dryer – 2024
- Evaluation of the drying characteristics of gotukola (*Centella asiatica*) and katurumurunga (*Sesbania grandiflora*) – 2024
- Feasibility of utilizing rice husk ash as a reinforcement material for soil-based roofing sheets – 2024
- Evaluation of an IoT-based infrared leaf temperature measuring system – 2024
- Design and development of a far-infrared specialty coffee roaster – 2022
- Feasibility of applying infrared radiation for drying cassava (*Manihot esculenta*) – 2022
- Maximizing the water condensate in the evaporator of a heat pump dryer – 2021
- Design parameters for an optimum batch-type heat pump drying process – 2021
- Feasibility of applying far-infrared radiation for drying turmeric – 2020
- Feedback control system with pulse-width modulation to control wavelength of infrared radiation – 2020
- Drying characteristics of coffee in an industrial-scale heat pump drying system – 2020
- Drying characteristics of paddy in a hot-air batch dryer – 2020
- Design and development of a two-wheel tractor-coupled bund plastering and canal (*kiwul-ela*) making equipment – 2019
- Design and development of a multi-crop hot air solar dryer – 2019
- Drying copra to extract high-quality coconut oil – 2014
- Kinetics of osmotic dehydration in coconut (*Cocos nucifera* L.) chips – 2014
- Assessment of potential and adoption of farm machineries by vegetable and potato cultivators in Nuwara Eliya District – 2013
- Design, development, and testing of a manually operated pineapple harvester – 2013
- Design, development, and testing of a small-scale pepper harvester – 2013

Scholarly Work

Research Impact Summary

Total journal publications: 23
Web of Science–indexed: 9 (Quartile distribution – SCImago Journal Rank: Q1: 3; Q2: 3; Q3: 2; ESCI: 1)
Scopus–indexed: 1
Other peer-reviewed: 13
Conference full papers: 11
Conference abstracts: 30
Book chapter: 1 (Springer Nature)
Patents: 1 granted; 2 filed; 1 submitted
Peer review service: 39 manuscripts

Journal Publications (Total: 23)

Web of Science–Indexed Journal Publications (SCI/SCIE/ESCI)

SJR Q1

Fernando, A. J. (2026). Artificial intelligence techniques for microwave drying of agricultural products: A comprehensive review on modeling, intelligent control, and process optimization. Applied Food Research, 6(1), 101590. `https://doi.org/10.1016/j.afres.2025.101590`. *IF 2024: 6.2, Clarivate JCR*

Fernando, A.J. and Rosentrater, K.A. (2023). Optimal designs of air-source heat pump dryers in the agro-food processing industry. *Food Engineering Reviews*, 15(2), 261–275. `https://doi.org/10.1007/s12393-023-09337-3`. *IF 2024: 7.6*

Fernando, A.J. and Amaratunga, K.S.P. (2022). Application of far-infrared radiation for sun-dried chili pepper (*Capsicum annuum* L.): Drying characteristics and color during roasting. *Journal of the Science of Food and Agriculture*, 102(9), 3781–3787. `https://doi.org/10.1002/jsfa.11726`. *IF 2024: 3.5*

SJR Q2

Fernando, A.J. and Rathnayake, R.M.S. (2025). Far-infrared radiation drying of cassava (*Manihot esculenta* Crantz) roots: Drying kinetics and quality attributes. *Journal of the ASABE*, 68(5). `https://doi.org/10.13031/ja.16287`. *IF: 1.0*

Fernando, A.J., Gunathunga, C., Brumm, T., and Amaratunga, S. (2021). Drying turmeric (*Curcuma longa* L.) using far-infrared radiation: Drying characteristics and process optimization. *Journal of Food Process Engineering*, 44(9). `https://doi.org/10.1111/jfpe.13780`. *IF 2024: 2.9*

Fernando, A.J., Amaratunga, K.S.P., Madushanka, H.T.N., and Jayaweera, H.R.Y.S. (2021).

Drying performance of coffee in a batch-type heat pump dryer. *Transactions of the ASABE*, 64(4), 1237–1245. `https://doi.org/10.13031/trans.14268`
(Note: Transactions of the ASABE is now published as the Journal of the ASABE.). IF: 1.0

SJR Q3

Fernando, A.J., Amaratunga, K.S.P., Dharmasena, D.A.N., and De Silva, T.H.K.R. (2025). Effect of temperature on drying kinetics and quality attributes of coffee beans during hot air drying. *Journal of the National Science Foundation of Sri Lanka*, 53 (3), 293 - 306. `DOI: 10.4038/jnsfsr.v53i3.12590`.

Kavindi, M.A.R., Amaratunga, K.S.P., Ekanayake, E.M.C., **Fernando, A.J.**, and Abesinghe, A.M.S.K. (2022). CFD simulation of airflow distribution in a heat pump-assisted deep-bed paddy dryer. *Applied Engineering in Agriculture*, 38(1), 1–8. `https://doi.org/10.13031/aea.14483`.

ESCI

Bandara, D.M.S.P., Amarathunga, K.S.P., Thilakaratne, B.M.K.S., Gunawardana, C.R., Dissanayake, T.M.R., Dharmasena, D.A.N., and **Fernando, A.J.** (2015). Use of evaporative water cooling in grinding chili. *Engineer: Journal of the Institution of Engineers, Sri Lanka*, 48(3), 13-17. `DOI:10.4038/engineer.v48i3.6837`.

Scopus-Indexed Journal Publications

SJR Q4

Chopra, S. and **Fernando, A.J.** (2020). Modeling employees' behavioral intention with the adoption of a suggestion system for lean initiatives. *Journal of Technology, Management and Applied Engineering*, 36(2), 1-14. `https://www.iastatedigitalpress.com/jtmae/article/id/14107/`

Other Indexed and Peer-Reviewed Journal Publications (DOAJ, Regional and International)

Loordhumary, A., **Fernando, J.**, and Geekiyanage, N. (2026). Effect of storage conditions on postharvest quality of water spinach (*Ipomoea aquatica* Forsk). *Advances in Technology.* 025, 5(3), 209-223 `DOI:10.31357/ait.v5i03.9102`

Fernando, A.J., Wimalaweera, T.P.M., Weerasooriya, G.V.T.V., and Bandara, D.M.S.P. (2025). Hot-air batch dryer for paddy: Evaluation of performance and quality of dried paddy. *Tropical Agricultural Research and Extension*, 28(3), 163–175. `DOI:10.4038/tare.v28i3.5771`.

Fernando, A.J. (2024). Transitions of food drying processes toward Industry 5.0. *Journal of Scientific and Engineering Research*, 11(8), 150–157.

Fernando, A.J. and Amaratunga, K.S.P. (2023). Psychrometric analysis of an open-loop batch-type heat pump drying process. *Asian Journal of Agriculture and Allied Science*, 6(1), 61–68.

`10.56557/ajaas/2023/v6i131`.

Fernando, A.J., Amaratunga, K.S.P., Dharmasena, D.A.N., Abeyrathna, R.M.R.D., Gajasinghe, I.L., Weerakoon, H.S.T., Ekanayake, E.M.A.C., and Bandara, D.M.S.P. (2022). Pulse-width-modulation control of a heat pump dryer with cascade evaporators and parallel-flow condensers. *Tropical Agricultural Research*, 33(1), 9–17. `DOI:10.4038/tar.v33i1.8532`.

Galappaththi, G.M.A.S., Weerasooriya, G.V.T.V., **Fernando, A.J.**, and Senanayaka, D.P. (2021). Development of solar-assisted multi-crop dryer. *Sri Lankan Journal of Agriculture and Ecosystems*, 3(1), 17–29. `DOI:10.4038/sljae.v3i1.58`.

Sandaruwan, E.A.A., Weerasooriya, G.V.T.V., and **Fernando, A.J.** (2019). Design and development of a two-wheel tractor-coupled bund plastering and canal (*kiwul-ela*) making equipment. *Sri Lankan Journal of Agriculture and Ecosystems*, 1(1), 102–116. `DOI:10.4038/cocos.v20i0.5796`.

Ekanayake, E.M.A.C., Amaratunga, K.S.P., Kariyawasam, H.K.P.P., **Fernando, A.J.**, and Abeyrathne, R.M.R.D. (2019). Design and development of a closed-cycle heat pump drying system for industrial drying of rice and chili. *Journal of Scientific and Engineering Research*, 6(12), 1–7.

Sewwandi, U.G.C., Ekanayake, E.M.A.C., Amaratunga, K.S.P., **Fernando, A.J.**, and Abeyrathne, R.M.R.D. (2019). Drying characteristics of *Katuwelbatu* (*Solanum virginianum* L.) during heat-pump drying. *International Journal of Scientific and Engineering Research*, 10(10), 1670–1672.

Fernando, A.J., Amaratunga, K.S.P., Priyadarshana, L.B.M.D.L., Galahitiyawa, D.D.K., and Karunasinghe, K.G.W.U. (2015). Roasting chilli (*Capsicum annuum* L.) using far-infrared radiation. *Tropical Agricultural Research*, 25(2), 180–187. `DOI:10.4038/tar.v25i2.8140`.

Bandara, D.M.S.P., Amarathunga, K.S.P., Thilakaratne, B.M.K.S., Dissanayake, T.M.R., Dharmasena, D.A.N., and **Fernando, A.J.** (2014). Feasibility study for evaporative cooling of grinding chili (*Capsicum annuum* L.). *Tropical Agricultural Research*, 26(1), 189–194. `DOI:10.4038/tar.v26i1.8083`.

Dissanayake, T. M. R., Amarathunga, K. S. P., Thilakaratne, B. M. K. S., Bandara, D. M. S. P., & **Fernando, A. J.** (2015). Gelatinization of rough rice using far-infrared (FIR) radiation. *Tropical Agricultural Research*, 26(4), 707–713. `DOI:10.4038/tar.v26i4.8133`.

Fernando, A.J., Adhikarinayake, T.B., and Weerasooriya, G.V.T.V. (2013). Design and development of a two-wheel tractor-driven coconut fertilizer applicator. *COCOS*, 20(1), 27–37. `DOI:10.4038/cocos.v20i0.5796`.

Conference Full Paper Publications (11)

Fernando, A.J., Mathapathi, S., Dharmasena, D.A.N., and Amaratunga, K.S.P. (2025). Non-Destructive Imaging for Okra (*Abelmoschus esculentus*) Maturity Assessment Using Machine Learning. In *Proceedings of the International Research Symposium on Postharvest Technology*, National Institute of Postharvest Management, Anuradhapura, Sri Lanka. [Accepted]

Karunanayake, M.S.H., Senanayake, S.M.V.P.D., Fernando, T.N., **Fernando, A.J.**, and Pramuditha, N.P.A.K.G. (2025). Evaluation of rice husk ash as a reinforcement material for the development of soil-based roofing sheet. [Accepted]

Fernando, A.J., Amaratunga, K.S.P., Gajasinghe, I.L., Weerakoon, H.S.T., Amarasinghe, A.A.P.S., and Wickramahewa, W.H.T.D. (2023). Development and thermal analysis of a multi-stage vapor compression heat pump dryer. In *Proceedings of the International Research Symposium on Postharvest Technology*, National Institute of Postharvest Management, Anuradhapura, Sri Lanka.

Fernando, A.J., Amaratunga, K.S.P., Weerakoon, S., Gajasinghe, I., and De Silva, R. (2021). Algorithm for calculating design parameters of a batch-type heat pump dryer. In *Proceedings of the 5th International Conference of the International Commission of Agricultural and Biosystems Engineering (CIGR)*, Canada.

Fernando, A.J., Amaratunga, K.S.P., Madhushanka, H.T.N., and Jayaweera, H.R.Y.S. (2020). Drying performance of coffee in a batch-type heat pump dryer. *ASABE Annual International Meeting*, USA.

Fernando, A.J., Amaratunga, K.S.P., and Priyadarshana, L.B.M.D.L. (2014). Far-infrared drying of stored coriander (*Coriandrum sativum* L.) up to grinding moisture content. In *Proceedings of the International Research Symposium on Postharvest Technology*, Institute of Postharvest Technology, Anuradhapura, Sri Lanka.

Fernando, A.J., Amaratunga, K.S.P., Priyadarshana, L.B.M.D.L., Galahitiyawa, D.D.K., and Karunasinghe, K.G.W.U. (2014). Far-infrared radiation roasting colour of chilli. In *Proceedings of IBA-IC: International Conference on India at the Crossroads – The Way Ahead.*

Panadura, S.L., Yalegama, L.L.W.C., and **Fernando, A.J.** (2014). Effect of drying efficiency on quality of coconut oil. In *Proceedings of the Special Sessions on Green Technology and Green Energy*, 5th International Conference on Sustainable Built Environment.

Zakeel, M.C.M. and **Fernando, A.J.** (2014). Learning style preference of a group of students attached to the Faculty of Agriculture, Rajarata University of Sri Lanka. In *Proceedings of the 10th Staff Development Centre, Sri Lanka Association for Improving Higher Education Effectiveness (SLAIHEE) Higher Education Conference*, 27–30.

Fernando, A. J., Ahikarinayake, T. B. and Weerasooriya, G. V. T. V. (2013). Development of a two wheel tractor driven coconut fertilizer applicator. Third annual research symposium, Rajarata University of Sri Lanka. 45. ISSN: 2235 9710.

Dassanayake, A. D. S. L., Bandara, M. H. M. A., Weerasooriya, G. V. T. V., and **Fernando, A. J.** (2012). Performance of a newly fabricated finger millet processing machine. Proceedings of the International symposium on Agriculture and Environment, University of Ruhuna. 261–263. ISSN: 1800-4830.

Conference Abstract Publications (30)

Samaranayake, B.D.T.R., **Fernando, A.J.**, Nilwala, N.G.N.A., and Jayaweera, J.M.D.M. (2025). Design and performance evaluation of an infrared dryer for paddy drying. *16th International Conference on Sustainable Built Environment (ICSBE 2025) – DIAMOND 75.*

Nilwala, N.G.N.A., **Fernando, A.J.**, and Samaranayake, B.D.T.R. (2025). Development of an IoT-based real-time monitoring and control system for a prototype infrared paddy dryer. *International Symposium on Agriculture and Environment (ISAE 2025).*

Karunanayake, M.S.H., Senanayake, S.M.V.P.D., Fernando, T.N., and **Fernando, A.J.**, and Pramuditha, N.P.A.K.G. (2025). Evaluation of rice husk ash as a reinforcement material for the development of soil-based roofing sheet. In Proceedings of the 3rd International Research Symposium (IERS 2025), National Engineering Research and Development Center, Sri Lanka.

Fernando, A.J. (2024). Food drying technology transitions towards Industry 5.0. In *First International Conference on University–Industry Collaborations for Sustainable Development (ICSD 2024).*

Sandaruwani, H.A.R., De Silva, A.B.G.C.J., **Fernando, A.J.**, and Medawatta, H.M.U.I. (2024). Evaluation of the drying characteristics of Gotukola (*Centella asiatica*) and Katurumurunga (*Sesbania grandiflora*). In *Proceedings of the 16th Annual Research Symposium*, Faculty of Agriculture, Rajarata University of Sri Lanka.

Karunanayake, M.S.H., Senanayake, S.M.V.P.D., Fernando, T.N., and **Fernando, A.J.** (2024). Feasibility of utilizing rice husk ash as a reinforcement material for soil-based roofing sheets. In *Proceedings of the 16th Annual Research Symposium*, Faculty of Agriculture, Rajarata University of Sri Lanka.

Hemantha, W.C., **Fernando, A.J.**, Weerathunga Arachchi, K., and Amarathunga, K.S.P. (2024). Performance evaluation of an industrial heat pump dryer. In *Proceedings of the 16th Annual Research Symposium*, Faculty of Agriculture, Rajarata University of Sri Lanka.

Loordhumary, A., **Fernando, A.J.**, and Geekiyanage, N. (2024). Effect of different wavelengths of light-emitting diodes on harvested water spinach (*Ipomoea aquatica* Forsk.). In *Proceedings of the 15th Annual Research Symposium*, Faculty of Agriculture, Rajarata University of Sri Lanka.

Weerakoon, W.M.M.D. and **Fernando, A.J.** (2024). Drying characteristics and quality of paddy exposed to infrared radiation. In *Proceedings of the 15th Annual Research Symposium*, Faculty of Agriculture, Rajarata University of Sri Lanka.

Wickramarachchi, M.R., Gunarathna, M.H.J.P., Geekiyanage, N., and **Fernando, A.J.** (2024). Evaluation of an IoT-based infrared leaf temperature measuring system. In *Proceedings of the 15th Annual Research Symposium*, Faculty of Agriculture, Rajarata University of Sri Lanka.

Loordhumary, A., **Fernando, A.J.**, and Geekiyanage, N. (2024, Nov 21–22). Application of light-emitting diodes (LED) on water spinach (*Ipomoea aquatica* Forsk). In *Proceedings of the International Conference on Innovation and Emerging Technologies 2024*, Faculty of Technology, University of Sri Jayewardenepura, Sri Lanka.

Rathnayake, R.M.S. and **Fernando, A.J.** (2022). Feasibility of applying infrared radiation for drying cassava (*Manihot esculenta*). In *Proceedings of the 14th Annual Research Symposium*, Faculty of Agriculture, Rajarata University of Sri Lanka.

Rathnayaka, R.K.A.B.L., Amaratunga, K.S.P., **Fernando, A.J.**, and Dissanayake, T.M.R. (2022). Design and development of far-infrared specialty coffee roaster. In *Proceedings of the Faculty of Agriculture Undergraduate Research Symposium*, University of Peradeniya.

Yapa, S.W.C.R.Y.M.S.P., Geekiyanage, N., Madhusanka, W.M.U.N., **Fernando, A.J.**, Gunathunga, C, N., and Rathnayake , R.A.A.S.(2022). Evaluation of an IoT–based smart soil warming apparatus for simulating varying degrees of soil temperature. Proceedings of the 14th Annual Research Symposium. Faculty of Agriculture, Rajarata University of Sri Lanka.

Fernando, A.J., Amaratunga, K.S.P., Dharmasena, D.A.N., Abeyrathna, R.M.R.D., Gajasinghe, I.L., Weerakoon, H.S.T., Ekanayake, E.M.A.C., and Bandara, D.M.S.P. (2021). Pulse-width-modulation control of heat pump drying through cascade evaporators and parallel-flow condensers. In *Proceedings of the 33rd PGIA Annual Congress*, Postgraduate Institute of Agriculture, University of Peradeniya, Sri Lanka.

Ekanayake, E.M.A.C., Amaratunga, K.S.P., **Fernando, A.J.** (2021). Modelling the particle size distribution of pulverized Gallnuts. In *Proceedings of the 33rd PGIA Annual Congress*, Postgraduate Institute of Agriculture, University of Peradeniya, Sri Lanka.

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Weerakoon, H.S.T., **Fernando, A.J.**, and Amaratunga, K.S.P. (2021). Design parameters for an optimum batch-type heat-pump drying process. In *Proceedings of the 13th Annual Research Symposium*, Faculty of Agriculture, Rajarata University of Sri Lanka.

Fernando, J. and Rosentrater, K. (2021, July 12–16). Overview of heat pump drying of agricultural products. *ASABE Annual International Meeting*, USA.

Jayaweera, H.R.Y.S., **Fernando, A.J.**, and Amaratunga, K.S.P. (2020). Feedback control system with pulse-width modulation to control wavelength of infrared radiation. In *Proceedings of the 12th Annual Research Symposium*, Faculty of Agriculture, Rajarata University of Sri Lanka.

Madushanka, H.T.N., **Fernando, A.J.**, and Amaratunga, K.S.P. (2020). Drying characteristics of coffee in an industrial-scale heat pump drying system. In *Proceedings of the 12th Annual Research Symposium*, Faculty of Agriculture, Rajarata University of Sri Lanka.

Wimalaweera, T.P.M., **Fernando, A.J.**, Weerasooriya, G.V.T.V., and Bandara, D.M.S.P. (2020). Drying characteristics of paddy in a hot-air batch dryer. In *Proceedings of the 12th Annual Research Symposium*, Faculty of Agriculture, Rajarata University of Sri Lanka.

Gunathunga, H.D.C.N.K., **Fernando, A.J.**, and Amaratunga, K.S.P. (2020). Feasibility of applying far-infrared radiation for drying turmeric. In *Proceedings of the 12th Annual Research Symposium*, Faculty of Agriculture, Rajarata University of Sri Lanka.

Galappaththi, G.M.A.S., Senanayaka, D.P., Weerasooriya, G.V.T.V., and **Fernando, A.J.** (2019). Design and development of a multi-crop hot air solar dryer. In *Proceedings of the 11th Annual Research Symposium*, Faculty of Agriculture, Rajarata University of Sri Lanka.

Sandaruwan, E.A.A.S., Weerasooriya, G.V.T.V., and **Fernando, A.J.** (2019). Design and development of a two-wheel tractor-coupled bund plastering and canal (*kiwul-ela*) making equipment. In *Proceedings of the 11th Annual Research Symposium*, Faculty of Agriculture, Rajarata University of Sri Lanka.

Fernando, A.J. and Chopra, S. (2016). Process improvement software platform for lean management in the health care sector. *ATMAE Annual Conference*, Association of Technology, Management, and Applied Engineering.

Dhanushika, D.D.I., Samaranayake, H.A.E., and **Fernando, A.J.** (2014). Kinetics of osmotic dehydration in coconut (*Cocos nucifera* L.) chips. In *Proceedings of the 6th Annual Research Symposium*, Faculty of Agriculture, Rajarata University of Sri Lanka.

Jayasiri, M.M.J.G.C.N., Weerasooriya, G.V.T.V., Kahandage, P.D., and **Fernando, A.J.** (2014). Design, development and testing of a manually operated pineapple harvester. In *Proceedings of the International Conference of Agricultural Science*, Faculty of Agricultural Sciences, Sabaragamuwa University of Sri Lanka.

Hapuarachchi, A.S.K., Weerasooriya, G.V.T.V., Kahandage, P.D., and **Fernando, A.J.** (2013). Design, development and testing of small-scale pepper harvester. In *Proceedings of the 5th Undergraduate Research Symposium*, Faculty of Agriculture, Rajarata University of Sri Lanka.

Dissanayaka, L.D.U.D.B., Bandara, M.H.M.A., Weerasooriya, G.V.T.V., Fernando, A.P.S., and **Fernando, A.J.** (2013). Assessment of potential and adoption of farm machinery by vegetable and potato cultivators in Nuwara Eliya District. In *Proceedings of the 5th Undergraduate Research Symposium*, Faculty of Agriculture, Rajarata University of Sri Lanka.

Thesis

Fernando, A. J. (2022). Optimization of heat pump drying through cascade evaporators and parallel-flow condensers. PhD thesis, Postgraduate Institute of Agriculture, University of Peradeniya

Fernando, A. J. (2015). Drying and roasting characteristics of chili (*Capsicum annum* L.) exposed to different wavelengths of far-infrared radiation. Master of Philosophy thesis, Postgraduate Institute of Agriculture, University of Peradeniya

Fernando, A. J. (2011). Design and development of a two wheel tractor driven coconut fertilizer applicator. Undergraduate thesis. Faculty of Agriculture, Rajarata University of Sri Lanka

Book Chapter

Fernando, A.J. (2026). Technological Transitions in Thermal Drying: Integrating Microwave and Heat Pump Technologies with AI and IoT to Enhance Productivity in Food and Agricultural Systems. In: Ahamed, T. (eds) *IoT and AI in Agriculture*. Springer, Singapore, pp. 311–341. `https://doi.org/10.1007/978-981-95-5218-4_21`

Karunanayake, M.S.H., Senanayake, S.M.V.P.D., Fernando, T.N., **Fernando, A.J.**, & Pramuditha, N.P.A.K.G. (2026). Evaluation of Rice Husk Ash as a Reinforcement Material for the Development of Soil-Based Roofing Sheet. In M. Siriwardhana, S. Kumar, M. Liyanage, J.G. Samarawickrama, I.U. Atthanayake, & S. Samarakoon (Eds.), *Selected Proceedings of the 3rd International Engineering Research Symposium (IERS 2025)* (pp. 186–195). *Proceedings in Technology Transfer*. Springer, Singapore. `https://doi.org/10.1007/978-981-95-7984-6_17`

Editorial and Peer Review Service

Peer reviewer for 42 manuscripts across 30 international journals, including several Q1- and Q2-ranked publications spanning agricultural technology, food and biosystems engineering, thermal process systems, and computational modeling.
Journals: Scientific Reports (Q1); Applied Thermal Engineering (Q1); Journal of Heat and Mass Transfer Research (Q2); Asia-Pacific Journal of Science and Technology (Q4); Food Science & Nutrition (Q1); Computers in Human Behavior (Q1); Journal of the Science of Food and Agriculture (Q1); Food and Bioprocess Technology (Q1); Cogent Food and Agriculture (Q2); Journal of Food Processing and Preservation (Q2); Agricultural Engineering International: CIGR Journal (Q4); Food Engineering Reviews (Q1); Sri Lankan Journal of Agriculture and Ecosystems; International Journal of Food Science and Technology (Q1); Biomass Conversion and Biorefinery (Q2); SN Applied Sciences; Engineering Computations (Q2); International Journal of Ambient Energy (Q2); Journal of Agriculture and Food Research (Q1); Current Research in Food Science (Q1); Journal of Biosystems Engineering (Q2); BMC Chemistry (Q2); Discover Applied Sciences (Q2); Computers and Electronics in Agriculture (Q1); Processes (Q2); Energies (Q1); Tropical Agricultural Research; Information Processing in Agriculture (Q1); Food Research International (Q1); International Journal of Refrigeration (Q1). Reviewer contributions (37) verified via Web of Science Reviewer Recognition Service.
Reviewed grants:
Reviewed conference papers:

Patents

- Sandaruwan, E.A.A., Weerasooriya, G.V.T.V., and **Fernando, A.J.** (2022). *A pull-type bund plastering and canal (kiwul-ela) making machine*. 20513. National Intellectual Property Office of Sri Lanka. (Granted)
- A computer program to generate the design parameters of an open-loop heat pump drying process - Application No: 22159 (Filed)
- A process of drying turmeric - Application No: 22158 (Filed)
- Design of a cascade evaporator heat pump dryer for crops (Submitted)

University, National, International, and Community Service

University and Faculty Service

- Facilitated the development of a Memorandum of Understanding between the People's Friendship University of Russia (RUDN University) and Rajarata University of Sri Lanka.
- Senate Member, Rajarata University of Sri Lanka (Faculty Nominee), 2022–2025.
- Coordinator, Faculty Business Incubation Center, 2024–2025.
- International Relations Coordinator, Department of Agricultural Engineering and Soil Science, 2022–2025.
- Panel member, Undergraduate Research Symposium of Technology, Faculty of Technology, RUSL, 2022, 2024.
- Faculty Coordinator, Quality Assurance Cell, 2024-2025.
- Academic Coordinator for developing the M.Sc. in Agro-Technology and Environmental Management, 2022–2025.
- Chairperson, Grace Chance Committee, 2024–2025.
- Member, Academic Development and Planning Committee, 2023–2025.
- Coordinator, Faculty Quality Assurance Cell for Managing the Learning Environment, 2023–2024.
- Student Club Adviser, Department of Agricultural Engineering and Soil Science, 2023–2025.
- Member, Postgraduate Curriculum Development Committee, 2024.
- Committee Member for developing By-Laws, Faculty of Agriculture, 2021–2022.
- Library Committee Member, Faculty of Agriculture, 2018–2022.
- Member, Program Review Document Preparation Committee, Faculty of Agriculture, 2018.
- Editorial Board Member, *Sri Lankan Journal of Agriculture and Ecosystems*, RUSL, 2023.
- Comprehensive Examination Panel Member, Master of Agriculture Degree Programme, 2022–2024.
- Developed online short courses: *Introduction to LaTeX* and *Introduction to Python Programming*, 2020.
- Online Examination Preparation Committee Member, 2021.
- Comphering in the inaugural session of the 31^{st} Postgraduate Institute of Agriculture (PGIA) congress, 2019
- Prepared Freshmen Student Handbook, Faculty of Agriculture, 2018–2022.
- Fund-Raising, Typesetting, and Food & Refreshment Committee Member, RUSL Annual Research Symposium, 2013.

National & International Contributions

- Evaluation of Detailed Application under Technology Development Grant Scheme (2025/TD/2/05) of National Science Foundation, Sri Lanka, 2025.
- Resource Person, Development of Grade 10 Agriculture Module, National Institute of Education, 2023.
- Faculty Coordinator, International Conference on Eco-Health Nexus: Bridging Cascade Ecology & Human Well-Being, 2023.
- Reviewer, International Postharvest Research Symposium, 2023.
- Judging Panel Member, *Sahasak Nimavum* – National Inventions and Innovations Competition, 2022.
- Reviewer for International Conference on Innovation and Emerging Technologies 2022.
- Attend Stakeholder Meeting on Performance Review of Farm Mechanization Training Center for 2016-2018, 2019.
- Reviewer, International Symposium on Agriculture and Environment, 2020.
- Co-controller, G.C.E. (A/L) Examination – Biosystems Technology (Practical), 2018.
- Session Moderator, Graduate and Professional Students Research Conference, Iowa State University, USA, 2016.

Community Engagement

- Sunday School Volunteer, 2019
- Organized farmer extension programs, 2012
- Served in the *Dayata Kirula* Exhibition, 2012
- Delivered Undergraduate Valedictory Speech, Faculty of Agriculture Research Symposium, RUSL, 2011

Faculty Research Symposium Involvement

Undergraduate Research Symposium (URS) and *Annual Research Symposium (ARS)*, Faculty of Agriculture, Rajarata University of Sri Lanka

- ARS 2024 – Evaluation Panel Member, 3MT Competition and Postgraduate Research Session; Member, Fund-Raising Committee; Editorial Board Member
- ARS 2022 – Member, Fund-Raising Committee
- ARS 2021 – Evaluation Panel Member, Farm Machinery Technical Session
- ARS 2020 – Member, Transport Arrangement Committee
- ARS 2019 – Department Coordinator; Member, Final Marks Processing, Medals, and Certification Committee

- ARS 2014 – Compere Coordinator
- URS 2013 – Secretary of the Symposium
- URS 2012 – Member, Organizing Committee

Workshops and Training Programs Organized/Conducted

- Workshop on Digital Tools for Teaching and Learning – 2024
- Awareness Session on Postgraduate Opportunities Abroad (Resource Person) – 2023
- Training Program on Using Assessment Blueprints and Rubrics for Academic Staff – 2023
- Training Program on Outcome-Based Education and Learner-Centered Teaching – 2023
- Stakeholder Workshop for Industrial Collaboration – 2014
- Motivational Workshop for Undergraduates on 'Ladder to Success' – 2014
- Soft Skills Development Workshop for Undergraduates – 2014
- Teacher Training Program, North Central Provincial Education Department – 2013
- Soft Skills Development Workshop for Undergraduates – 2013

Curriculum and Course Development

- **Postgraduate Curriculum Development:** Member, Curriculum Committee for introducing the new course *Advanced Machinery Systems Engineering* to the Master of Agriculture Degree Program, RUSL.
  Contributed to the development of new courses for the proposed *M.Sc. in Agro-Technology and Environmental Management*: *Innovation and Product Development, Agro-Product Processing Technology, Research Methods and Scientific Communication, Electronics and Instrumentation in Agricultural Technology, IoT and Big Data Analytics for Agriculture and Environment,* and *Thesis/Research Project.*
- **Undergraduate Curriculum Development:** Member, Curriculum Development Committee (2014–2018); contributed to introducing new undergraduate courses *Precision Agriculture* and *Workshop Engineering.*
- **Course and Lesson Plan Revisions:** Introduced and revised lesson plans for the 2021–2025 undergraduate curriculum, including *Thermodynamics, Food Process Engineering, Engineering Physics,* and *Innovation and Product Development.*

Additional Information

Extracurricular Achievements – Undergraduate Level

- Member, Quiz Competition – First Runner-up, 2010.
- Member, Debate Team (Senior Level) – Runner-up, 2010.
- Member, Debate Team – Participant, 2009.

- Member, Faculty of Agriculture *Elle* (Women's) Team – Secured Third Place in the Inter-Faculty Tournament, 2008.

Early Leadership & Extracurricular Involvement – School Level (Ave Maria Convent, Negombo)

- Senior Prefect, 2004
- Junior Prefect, 2002
- President – Social Studies Question Bank
- Vice Secretary – English Literary Association
- Member – School Eastern Band
- Member – Netball Team

Certifications – Speech & Communication

- Merit Award – Verse Speaking, The Second Sri Lankan Annual Competitive Festival of Music, Speech, Drama, and Dance, Negombo (affiliated with the British Federation of Festivals, UK)
- Trinity College London – Speech – Distinction

Personal Interests

Gardening and international travel – have visited Australia, the United States, Japan, France, Italy, India, Thailand, Portugal, Austria, Slovakia, the Czech Republic, Poland, and Hungary

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